

Divya Pandey

Research Scientist

A result-driven research professional with 6+ years of experience in the areas of Microbiology, Molecular Biology, Nanotechnology, and Immunology. Currently with Catalysts Biotechnologies Pvt. Ltd as Research scientist (since Dec 2020) leading Microbiology department. Specialized in working in a fast-paced environment and liaison between different stakeholders to plan and implement research projects. Worked in Microbiology dept., All India Institute of Medical Sciences (AIIMS), Delhi as Research Assistant and an ICMR-individual Senior Research Fellow. Worked with Wrig Nanosystems Pvt. Ltd.(July 2016 – Feb 2017) as Research Associate in biomedical devices fabrication and validation projects using The Metrohm Autolab. Well versed in scientific tools and techniques, aerobic/anaerobic microbial isolation and identification using conventional methods and automated techniques (MALDI TOF and VITEC), MLT, Sterility testing, surface swab testing, Environment biomonitoring, fumigation, DNA and RNA extraction, PCR using thermal cycler, Gel electrophoresis, Gel Doc and UV trans-illuminator, Minimum inhibitory concentration (MIC) study using high throughput 96 well plate-MTP reader and UV-Vis spectrophotometer, ethanol fermentation, SDS PAGE, and ELISA (quantitative and qualitative), FTIR spectroscopy, microscopic techniques SEM, TEM and AFM. Also handled ISO 2015: 9001 and NABL audit documentation.

Contact

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Skills

- **Microbial identification/characterization**
- **Nucleic acid testing (NAT)**
- **ELISA**
- **Audit Documentation**
- **Statistical analysis**
- **Scientific writing**

Education

2014 – 2016

**M.TECH Industrial Biotechnology
Delhi Technological University
CGPA: 8.45/10**

2008 – 2012

**B. TECH Biotechnology Amity
University
76.22%**

2008

**(CBSE) Senior Secondary:
Science (PCB) Navayuga
Radiance, Lucknow
84.80%**

2006

**(CBSE) Higher Secondary
Navayuga Radiance, Lucknow
74.22%**

Professional Experience

Research Scientist-Microbiology (Nov 2020 – Nov 2023) Catalysts Biotechnologies Pvt. Ltd.

- Lead Microbiology and Molecular biology division.
- Project delivery on time, within agreed budget and in a quality satisfactory to the client.

Major Projects Handled:

- Press-mud treatment for enhancing biogas production in anaerobic digester using enzyme-based formulations. (Ongoing)
- Development of antimicrobial products for increasing the fermentation efficiency to achieve desirable ethanol yield at distilleries. (Completed with industrial trial)
- Development of biocides-based formulation for microbial growth control in the milling sections of Sugar Industry. (Completed with industrial trial)
- Development of high gravity yeast for enhancement of molasses ethanol productivity during fermentation in molasses/grain industry. (Completed-strain deposited in IMTECH, Chandigarh. Indian patent granted for HG.)
- Development of algacide for algae control in cooling towers and other water handling equipment in the industry. (Completed with industrial trial)
- Screening, Isolation, purification, characterization and preservation of microbial flora present in different industrial samples of molasses/grain.

Technical expertise:

- Microbial analysis: total bacterial count, Total Lactobacillus Count, Total Yeast and Mold Count using pour plate and spread plate technique, Yeast Cell Count using hemocytometer, Pathogen testing, Minimum inhibitory concentration (MIC) study using MTPR unit, Environment Bio-monitoring, Microbial limit test, Sterilization using autoclave using chemical strips, Fumigation, surface swab testing, ethanol fermentation and distillation.
- DNA extraction, nucleic acid amplification and quantification, agarose gel electrophoresis, SDS-PAGE etc.

Responsibilities:

- Self-directed trouble shooting with available resources and information.
- Diligent in following GLP guidelines.
- Documentation for internal and external (ISO 2015: 9001) audits.
- Writing research articles and scientific communications.
- Work closely with clients and marketing teams to provide scientific back-up and new insights.

Senior Research Fellow (Jul 2019 - Nov 2020)

ICMR-Individual Fellow, SBPL, Dept. of Microbiology, AIIMS, Delhi

- Worked on project entitled as: A Sequential Assessment of the Gut Microbiome in the patient of Inflammatory Bowel Disease by Next Generation Sequencing and Metagenomics

Research Assistant—Clinical Trial (Feb 2017- Jul 2019)

SBPL, Microbiology dept., AIIMS, Delhi

- Project worked on "A study to assess the effect of a probiotic strains on gut microbiota and mucosal immunity in Inflammatory Bowel disease. Trial reg.: Clinical Trials Registry India (CTRI) Identifier: Ref CTRI/2019/11/022087, REF/2016/09/012181.
- Technique involved culturing of aerobic/anaerobic microbial colonies and their identification, antimicrobial effectiveness test, DNA extraction from stool and blood samples, Polymerase chain reaction (PCR), gene sequencing analysis of outsourced Next Generation Sequencing (NGS), quantitative Enzyme Linked Immunosorbent Assay (ELISA) for pro and anti-inflammatory markers etc.
- Enrollment of IBD patients from OPD. Examination and analysis of Psychological and Physiological parameters using designed questionnaire as per the Hopkins Symptom Checklist (HSCL)
- Handling of sophisticated instruments like Anaerobic gloves box, digital microscope, VITEC, thermal cycler, Electrophoresis unit, Gel Doc, Tecan Nanodrop for DNA quantification, ultracentrifuge, etc.
- Drafting of project proposal for ICMR individual research fellow
- Documentation related to project report, ethical approvals, and manuscript drafting

Research Associate (Jul 2016-Feb 2017)

Wrig Nanosystems Pvt. Ltd., New Delhi

- Designing, development, optimization and validation of glucometer sensing device (Amperometry/Potentiometry)
- Validation of developed blood Hemoglobin detection using TrueHb meter
- Vendor management

Academic and Professional Achievements

- Have received "Special Appreciation Award, for Microbiology team" as Research Scientist in Catalysts Biotechnologies. (11 April, 2023)
- Have received "Star Achiever Award," as Associate Research Scientist in Catalysts Biotechnologies. (16 April, 2022)
- An oral presentation during the 79th Annual Convention of STAI on Microbial Diversity in Molasses of Northern and Western region of India." (11 Oct, 2021)
- Have received the ICMR individual Senior Research fellowship award for my shortlisted project. (August 2019)
- Presented a poster entitled as "A Search for Microbial Bio-markers for Colon Cancer in Indian Population using a Next Generation Sequencing Approach during the conference "9th India Probiotics Symposium," Probiotics through the lifespan," (25 Nov, 2018)
- Received "Young investigator Award" for presenting a research work during the conference "4th Biennial Conference of PAI and International symposium on Probiotics Therapy: Translating to Health and Clinical Practice." (2018)
- Presented the poster entitled as "Protein functionalized nano-structured zirconia based electrochemical immunosensor for cardiac troponin I detection," during the conference organized by OMICS group (2015)
- Participated in the first international conference on "Recent Advances in Bio-Energy Research at SSS-National Institute of Renewable Energy, Jalandhar. (2015)
- Attended three days national training on "Practical Hands for bio-fuel production and characterization from non- edible vegetable oils." at SSS-National Institute of Renewable Energy, Jalandhar. (2015)
- Secured AIR 386 rank with 525 score in GATE-2014 and qualified GATE-2015.

Academic Projects

- Fabrication of surface plasmon resonance biosensor based on graphene wrapped cysteine self-assembly for cardiac troponin I detection. (Nanobioelectronic Lab, DTU, 2016)
 - Synthesis of nanocomposites & self-assembled monolayers, immobilization of enzymes, characterization using contact angle, FTIR, SEM and TEM, optimization using Surface Plasmon Resonance (SPR), KEI Twingle and data evaluation using Kinetic evaluation software.
- Protein functionalized nanostructured zirconia based electrochemical immunosensor for cardiac troponin I detection. (Nanobioelectronic Lab, DTU, 2015)
 - Synthesis and functionalization of nanomaterials, electrophoretic deposition (EPD) of nanomaterials, fabrication of bio-electrode, characterization using SEM, TEM, AFM, FTIR, determination of sensitivity,

- stability, shelf life of the biosensor using Metrohm Autolab, The Netherlands, data evaluation using Origin Pro software.
- Optimization, production, and characterization of bio-surfactant producing microbes isolated from soil sample and expression of bio-surfactant producing gene in *E. coli*. (MRD Life sciences, Lucknow, 2012)
 - Serial dilution of the soil samples, culturing of samples, microscopy, DNA amplification via PCR technique
- Control testing of medicines manufactured in Cadila pharmaceutical private (Ltd), Ahmedabad
 - Sterility test, microbial limit test, Microbial enumeration test, and Bacterial endotoxin test etc.
- Qualitative and Microbiological Testing Techniques used in Analysis of Milk and Milk products from Parag, Lucknow
 - Processing of milk and milk products, Pasteurization, testing using lactometer, microbial culture, microbial limit test, stability testing, Packaging of milk and milk products etc.
- Transformation and Screening of Bacterial Strain DH5α using Recombinant DNA Technology Techniques from Allele Life sciences, Noida, U.P.
 - pDNA isolation, gDNA isolation, electrophoresis, transformation, blue-white screening, buffer preparation, centrifugation, sterilization using autoclave, working under Laminar Air Flow (LAF), handling of microbe.

Published Research articles and e-proceedings

- Anil Kumar Rai, Divya Pandey and K.V.T.S Pavan Kumar Sugarcane Syrup Preservation - A Combinatorial Approach. E-Proceedings of 81st Annual Convention of STAI. September 2023.
- Bamola, V. Deepak, Divya Dubey, Projoyita Samanta, Saurabh Kedia, Vineet Ahuja, Ratna Sudha Madempudi, Jayanthi Neelamraju, and Rama Chaudhry. "Role of a Probiotic Strain in the Modulation of Gut Microbiota and Cytokines in Inflammatory Bowel Disease." (2022). *Anaerobe*. 02 Oct. 2022. <https://doi.org/10.1016/j.anaerobe.2022.102652>
- B. Chandrashekhara, S. Shrotriya, Divya Pandey and K.V.T.S Pavan Kumar. Improvement of Anaerobic Treatment of Distillery MEE Condensate by Adding Nutrient Supplements. E-Proceedings of 79th Annual Convention of STAI 11. July 2022.
- Divya Pandey, M. Madan, S. Kushwaha and K.V.T.S. Pavan Kumar et al. "Microbial Diversity in Molasses of Northern and Western region of India." e-Proceedings of 79th Annual Convention of STAI 11. October 2021
- Bamola, V. Deepak, Divya Dubey, Projoyita Samanta, Saurabh Kedia, Ratna Sudha Madempudi, Jayanthi Neelamraju, Vineet Ahuja, Rama Chaudhry. Effect of *Bacillus coagulans* Unique IS-2 in Inflammatory Bowel Disease (IBD): A Randomized Controlled Trial. DOI: <https://doi.org/10.1101/2021.07.18.21260556>
- Chaudhry, Rama, Vishwa Deepak Bamola, Projoyita Samanta, Divya Dubey, Tej Bahadur, Monica Chandan, Shyam Tiwary et al. "Immunoglobulin receptors expression in Indian colon cancer patients and healthy subjects using a non-invasive approach and flowcytometry." *International Journal of Applied and Basic Medical Research* 10, no. 3 (2020): 194. DOI: [10.4103/ijabmr.IJABMR_191_19](https://doi.org/10.4103/ijabmr.IJABMR_191_19)